An ABM on Covid-19 Transmission Dynamics

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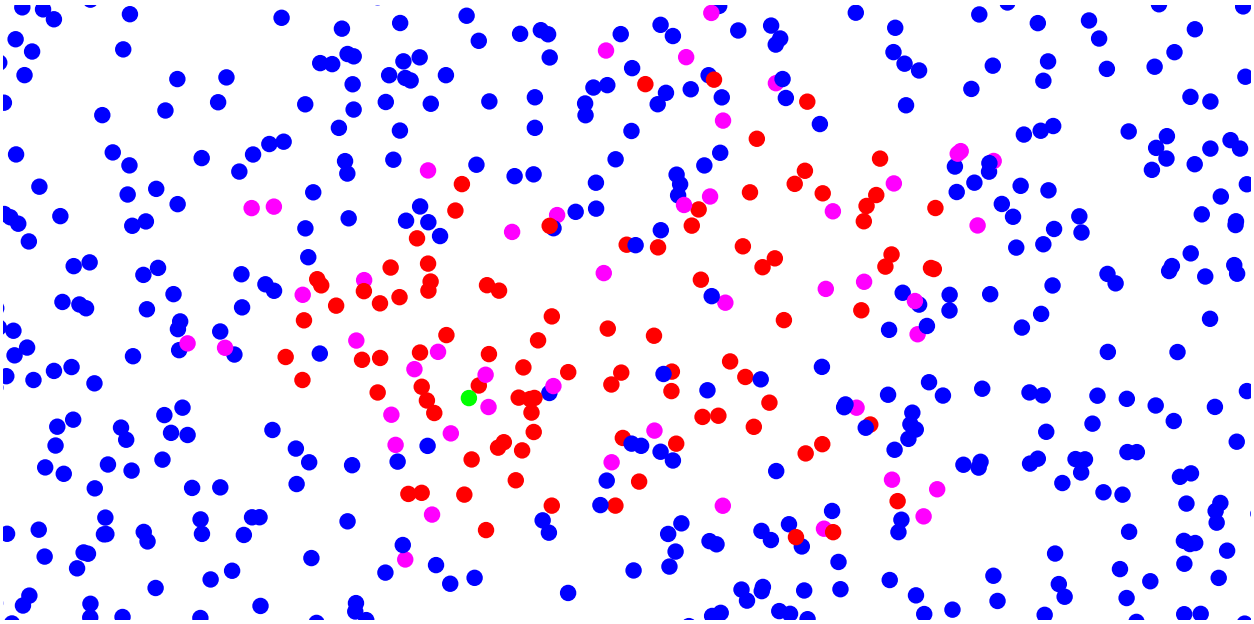


Abstract

Studies have been conducted about the transmission of Covid-19 (). However, the interactions of the different transmission factors with space and mobility have not been throughly investigated. This study aims to investigate the effects of the factors of social approval, quarantine, and vaccination on Covid-19 transmission. To study such effects in a more realistic setting, it uses agent-based modeling to incorporate space and mobility. As the result, it shows that longer testing intervals allows more agents tested positively to synchronize their quarantine, which could efficiently end the transmission for all health conditions early and control the transmission. Also, the effects of higher social penalty and vaccination coverages could be less salient once their saturation points were reached. While Covid-19 transmission is a complex system, conducting this type of research under space and mobility can provide more insights into the mechanisms of the interactions of the different casual factors, which allow us to better prepare and response to the similar situations in the future.

1. Introduction

In this paper, the effects of testing, social approvals, and vaccinations on the Covid-19 transmissions in US were systematically investigated. AMB simulations were used to answer the research questions of: What are the effects of the different testing intervals and quarantine lengths? What role do social approvals play in the transmission dynamics during the pandemic? How would the different vaccination coverages and restrictions affect the population conditions over time? By studying such questions, it could shed more light on the impacts and the epidemiology mechanisms of this pandemic. To make a realistic epidemiology model, it included some different factors, such as age, mobility, household transmissions, mask wearing, supplement usage, and virus mutation. Through the interactions of these factors, it also allows us to better understand and analyze the virus transmission and its impacts in population during the pandemic. By applying the results of this model, it can also allow us to make better predictions and responses for the similar transmissions in the future, which can also help to reduce the potential loss and health risks of the population. For instance, this model found that social approvals had a positive effects in encouraging mask wearing and lower the transmissions. This could help to explain why some countries that emphasize social approvals were able to resolve the pandemic more efficiently. By strengthening people's awareness of mask wearing and addressing the significance of social cooperations in the future, it could lead to a better result in controlling the spread of the disease. Furthermore, the model also found the effects of the increased testing interval, quarantine lengths, social penalty, and vaccination coverages beyond certain points would have less effects in the population. Thus, by relocating the excess resources from such categories into incorporating more testings and other facilities might lead to a more efficient resource distribution and better outcome. Although different research works have been done on the effects of the different vaccines, testing distributions, and social behaviors on the Covid-19 transmission, many of them were mostly focused on the mathematical aspects of the analysis and few have incorporated the space mobiles or the effects of social approvals inside the transmission dynamics (Komaroval et al., 2020, M. K. et al., 2020, Komarova et al., 2020). However, in this paper, it used ABM to incorporate the different transmissions factors with mobile agents to investigate the their influences in the real space, which could provide more insights into the spread of this disease and the nature of its epidemiology.

2. Methods (model description)

In this model, mobile agents were placed on the real-life coordinates in a 200x200 space. At the start of the simulation, agents were randomly assigned an age from the Gaussian distribution with an average age of 150 steps and standard deviation of 0.1. Based on the proportion to human lifespan, this model assumes the max age was 230 steps (UNHCR, 2018),

and defines the "elder age" with 123 steps (Thompson, 2018). They were also randomly assigned to wear masks, be a part of a household, or take supplements. In the simulation, the agents moved toward random directions in the space. To simulate the transmission progression, this model used 5 health conditions: Susceptible, Exposed, Infectious, Recovered, and Dead. In the beginning of each simulation, 7000 susceptible and 5 infected agents were randomly placed in the space. At each step, if an agent has an infectious neighbor within 6 feet (CDC, 2020), its risks of being transmitted with the virus would be evaluated based on the factors of its transmission group, age, supplement usage, mask wearing, vaccination condition, and virus mutation. If an agent or its infectious neighbor didn't belong to the household group, it would receive a 0.09 for its group transmission probability(GTP) (Worldometers, 2021). But if both of them were in the household group, its GTP would be 0.17 (Fung, 2020). At the evaluation stage, an agent started with an initial current transmission probability (CTP) equals to its GTP, but its actual transmission probability (ATP) would be its final CTP that was influenced by the other mentioned factors. For an elder agent, its CTP would be its GTP times 0.88 (Davies, 2020). For an younger agent, it would be its GTP times 0.62 (Davies, 2020). If an agent takes supplements, its risk would be lowered (Grant, 2020), and it would receive a 12% reduction in its CTP (Kingsland, 2020). If it wears mask, it would receive a 70% reduction in its CTP. If it was vaccinated, its ATP would be set to 0. However, if a virus mutation was introduced, there would be a mutation rate of 0.001 (Drake, 1998). After a mutation, a agent would receive a 50% increase in its CTP (CDC, 2021). After the evaluation of these factors, an agent would become exposed and start its transmission countdown of 2 steps (CDC, 2020). When this countdown was ended, it would become infectious and start a countdown of 10 steps for its Infectious stage (CSC, 2021). When this countdown was over, it would progress to the Recovered or Dead conditions, based on its age. For instance, an elder agent would have a 0.807 dead rate (CDC, 2021), and a younger agent would have a 0.097 death rate (CDC, 2021). If an agent was dead, it would be removed from the space. However, if it didn't die in this evaluation, it would be recovered and begin its countdown of 180 steps for this stage (CDC, 2021). Once its countdown for its Recovered phase was ended, it would become susceptible again. For all simulations, the health conditions for the agents were checked at each step, and the progression of their health conditions remained the same. In the simulations, each step corresponded to one day in the real-world. Furthermore, the simulations were run with a length of 550 steps, and each of them was run 10 times with a sampling interval of 7 steps.

3. Results

3.1 Simulation 1: Basic model

The basic model simulates the effects of the household transmission on the progression of the population health conditions with no other restrictions. In a typical run of this simulation, the percentage of the exposed agents remains the highest during the first 90 steps, followed by that of the infectious, recovered, and dead (Fig. 1(a)). This could be explained by the higher transmission rate of the household groups in the population. It's also consistent with the data of

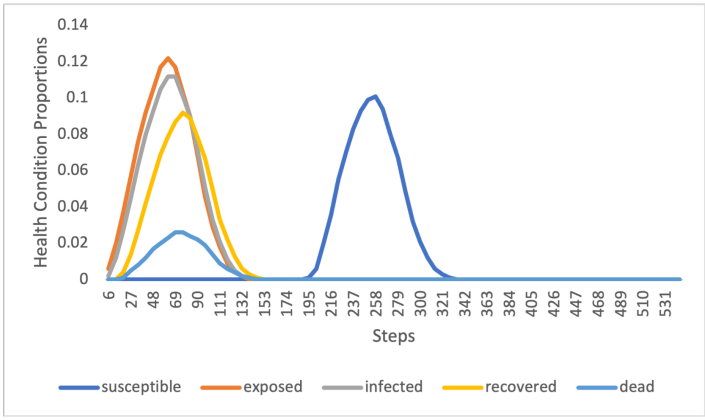


Fig. 1(a). A basic model with no other restrictions besides checking the health conditions at each step.

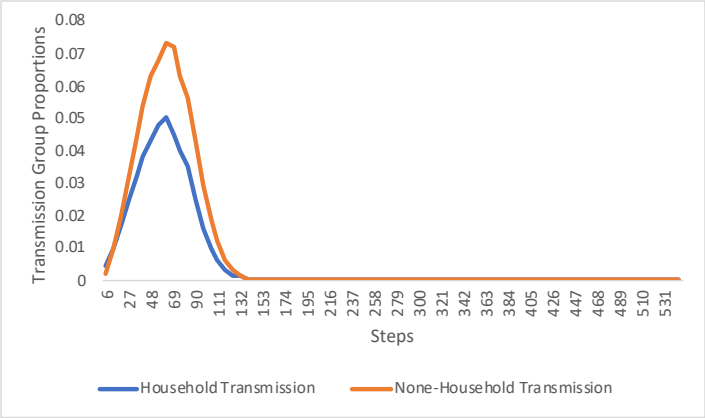
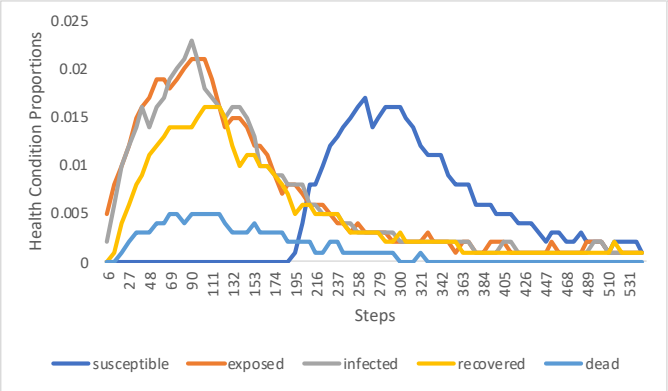


Fig. 1(b). The proportions of the household transmissions and none-household transmission of the basic model

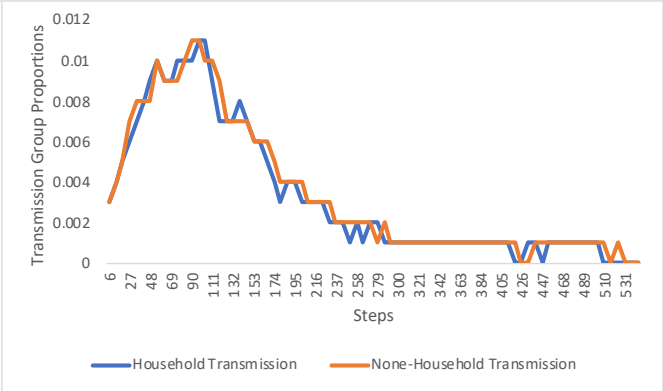
the transmission proportions, where the percentage of the household transmission reaches its peak (0.05) around the 69th step and it was almost 70% of that of the none-household transmission (0.072) (Fig. 1(b)). However, the death proportion was also the highest (0.026) at around the 69th step. After 153 steps, all agents were recovered. The recovered agents began to become susceptible again at around 195th step, which created a wave of susceptible agents between the 195th and 342th step.

3.2 *Simulation 2: Testing & Quarantine*

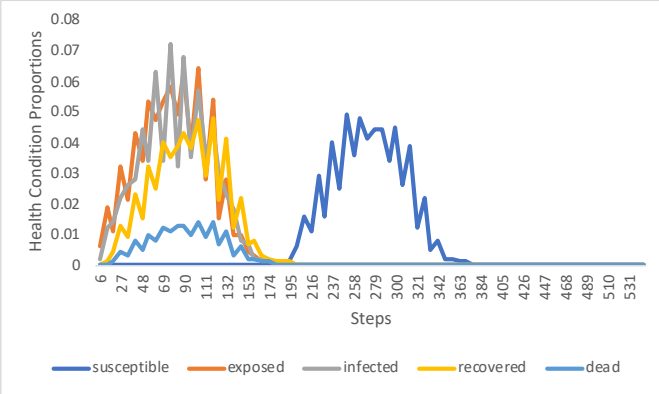
The second simulation studied the effects of testing and quarantine in the agent population. In this simulation, it assumes all agents were able to be tested when their testing countdown ended. After each testing interval, the agents received their Covid-19 tests. If an agent was tested positive, it would begin its quarantine process. For the course of its quarantine, the agent would stay at its current position in the space and could no longer expose or become exposed with the virus. In the simulation, the combinational effects of the different testing frequencies (7, 14, 21) and quarantine lengths (14, 21) were studied. Overall, all forms of testing and quarantine led to the reductions of proportions of all conditions. However, the combination of the testing frequency of 7 steps and quarantine length of 14 steps appeared to have the most effective impact on reducing the transmission proportions (Fig. 2.1(a)). Compares to the basic model (Fig. 1), this combination shows a substantial drop in the proportions of all conditions. Namely, the peak of the exposed proportion went from 0.12 to 0.021, which was a 82% decrease. This could be easily contributed by the relatively frequent testing conditions, since the more frequent testings can discover and quarantine the infectious agents more efficiently. And its effect could also be reflected from (Fig. 2.1(b)), where the proportion of household and none-household transmission both greatly dropped and remained the same on average. For the trials with the same quarantine length of 14 steps but had longer testing frequencies of 14 and 21 steps, the magnitudes of the oscillations for all conditions increased with longer testing frequencies



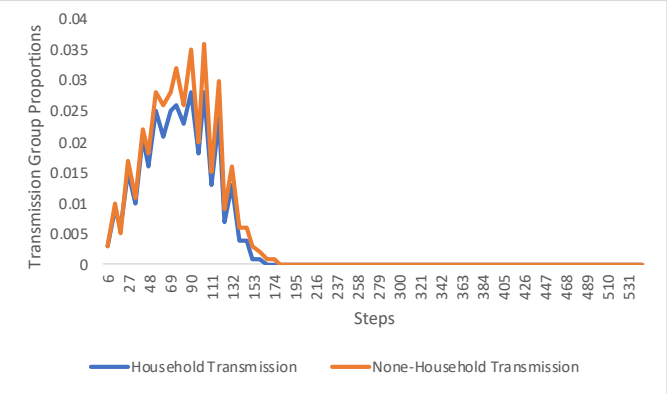
(a) testing interval=7, quarantine length=14



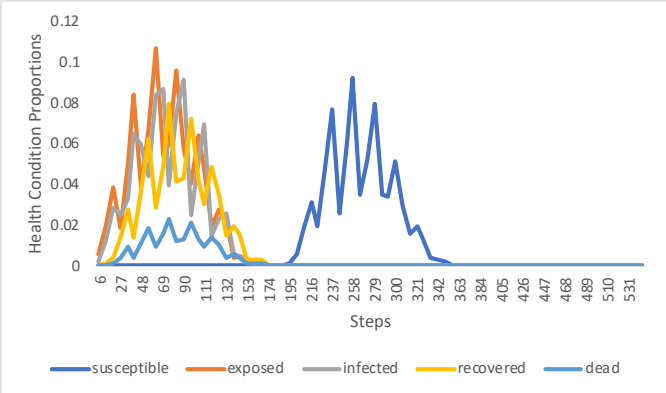
(b) testing interval=7, quarantine length=14



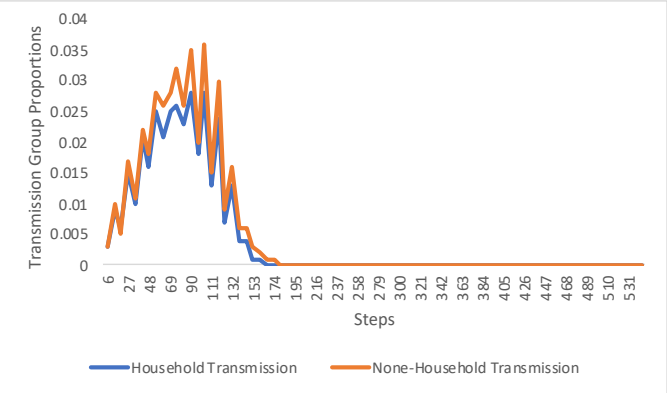
(c) testing interval=14, quarantine length=14



(e) testing interval=14, quarantine length=14



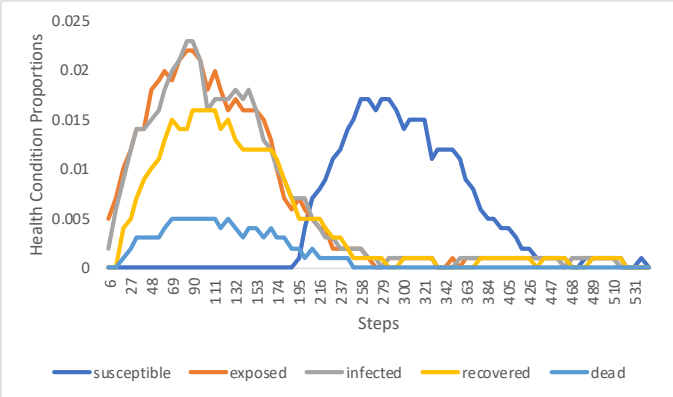
(f) testing interval=21, quarantine length=14



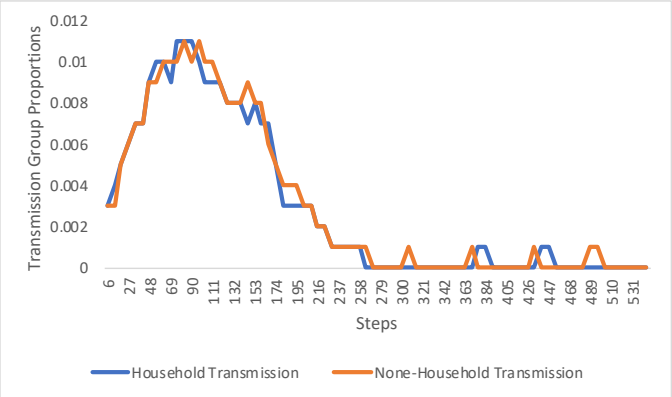
(g) testing interval=21, quarantine length=14

Fig. 2.1. These charts depict the health condition and transmission group proportions under the testing intervals of 7, 14, 21 and the quarantine length of 14

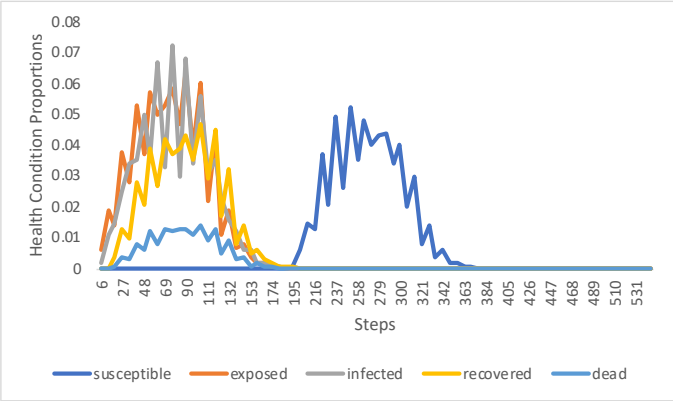
(Fig. 2.1) This could be because that longer testing frequencies can create different gaps between the times where agents were tested and were waiting to be tested, which created a less smooth proportion graph. Although the effects of the combination with longer testing frequencies were not as robust as the combination 1 in Fig. 2.1(a)(b), their proportions of all conditions and group



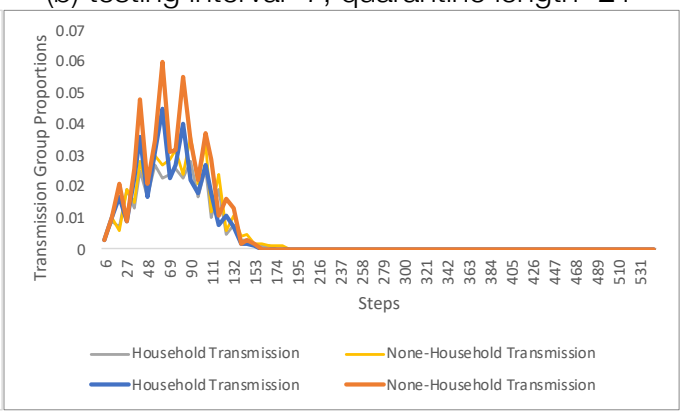
(a) testing interval=7, quarantine length=21



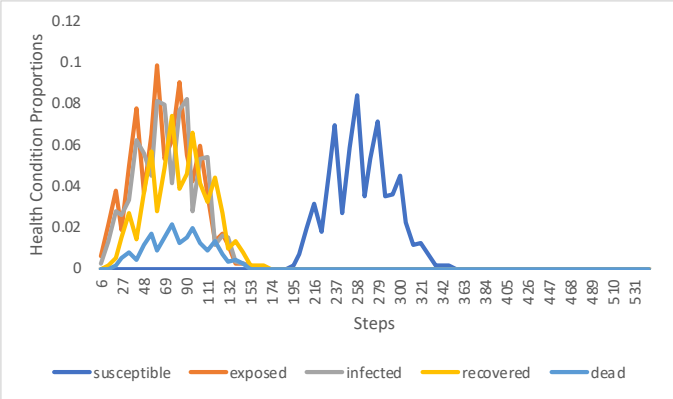
(b) testing interval=7, quarantine length=21



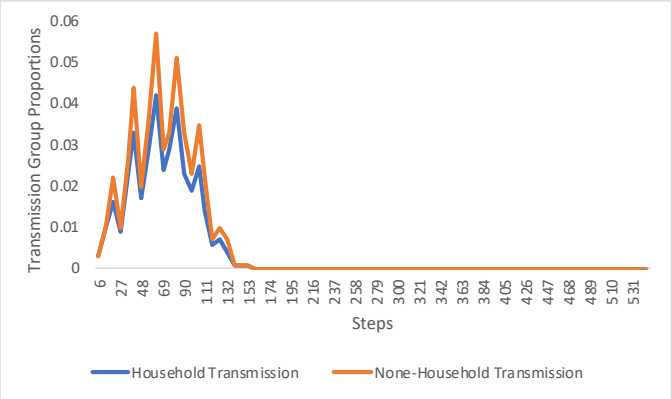
(c) testing interval=14, quarantine length=21



(d) testing interval=14, quarantine length=21



(e) testing interval=21, quarantine length=21



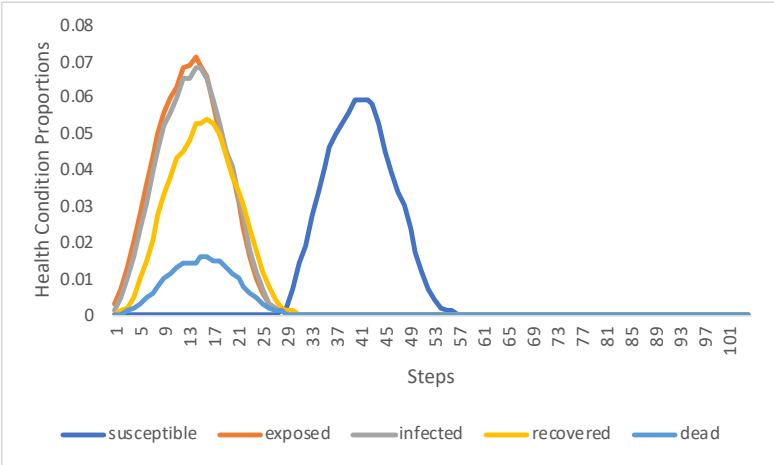
(f) testing interval=21, quarantine length=21

Fig. 2.2. The charts depict the health condition and transmission group proportions under the testing intervals of 7, 14, 21 and the quarantine length of 21

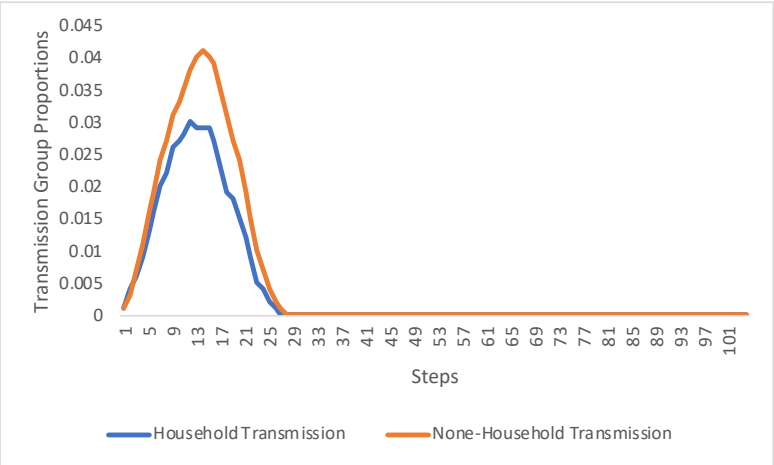
transmissions reached 0 after 384, 174 steps and 356, 153 steps respectively. Compares to the >545 and 531 steps of the combination 1, these trials were able to end much sooner. The reason for this might be because that the longer testing intervals could allow more agents who became

tested positive to start their quarantine at once, which reduced the ending time for all conditions and transmissions more efficiently. In comparison, although the combination 1 had a higher testing frequency of 7 steps, it might had fewer agents tested positive after each 7 steps, which could lead to a longer time period for all infectious agents to be tested and quarantined. However, under the quarantine length of 21 steps, the combination in Fig. 2.2(a)(b) had the similar effects in conditions and group transmissions as combination 1, but combination 1 still remained as the most effective one, since it had more subtle drops in condition proportions and steeper decreasing slopes on average. For the ones with the increased testing interval of 14 and 21, they also had the similar oscillations and early ending date for all conditions and group transmission as their corresponding combinations under the quarantine length of 14 steps (Fig. 2.2). In the graphs, it could also be seen that longer testing intervals and quarantine lengths had fewer effects in further reducing any conditions or group transmissions. This could be because that some of the effects of the longer quarantine lengths were counteracted by the longer testing frequencies, which could lead to a smaller effects in transmission reduction.

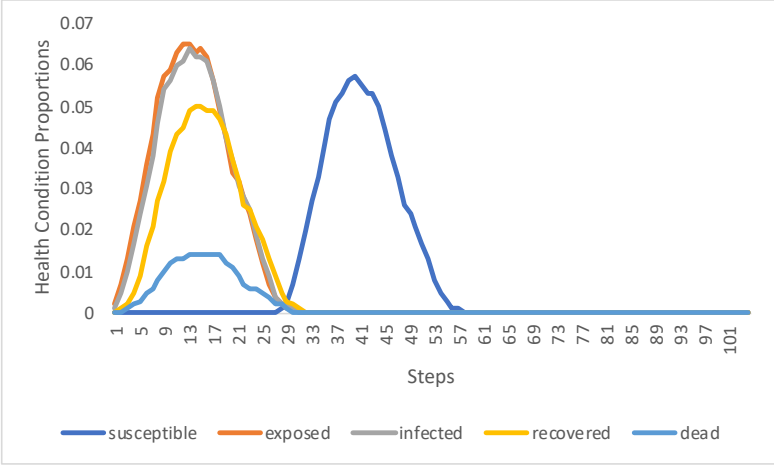
3.3.1 Simulation 3: Social approvals



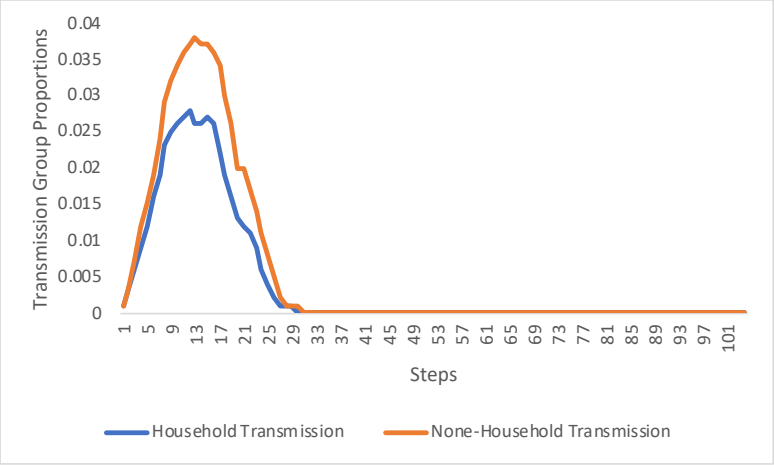
(a) Social Penalty=10



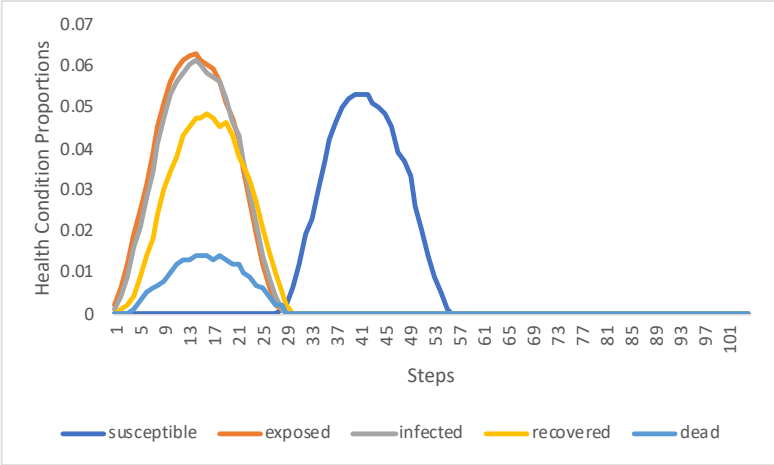
(b) Social Penalty=10



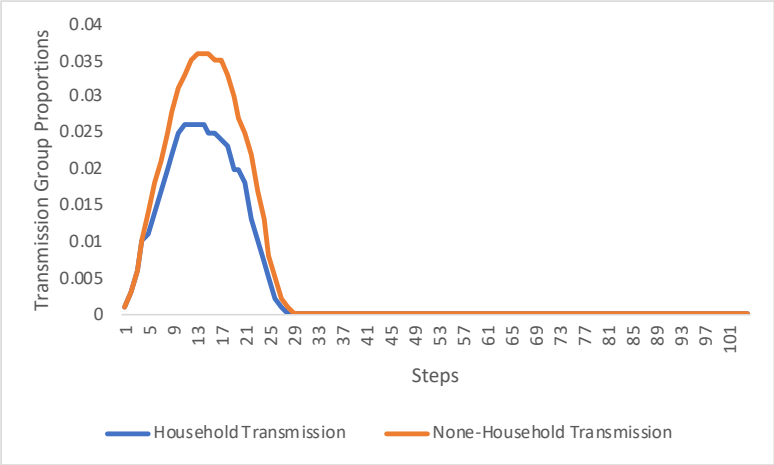
(c) Social Penalty=20



(d) Social Penalty=20



(e) Social Penalty=30



(f) Social Penalty=30

Fig. 3. As the social penalty increased, the different health conditions and their group transmissions were effectively reduced.

In this simulation, it investigated the effects of social approval in encouraging mask-wearings and the reduction of virus transmissions. At the beginning of the simulation, all agents started with an approval score of 100, and they were randomly assigned to wear masks or not wear masks. During the simulation, when an agent met one of its neighbors within 6 feet, one of them from this dyad would receive a social penalty (approval score reduction) if it didn't wear a mask but the another one did. To better investigate the effects of this factor, different social penalties of 10, 20, and 30 were studied. At each step, the approval scores of the agents were checked. If an agent's approval score dropped below the approval baseline of 70 and it didn't wear a mask, it would put its mask on to restore its initial score of 100. In this simulation, it's assumed that the agents would not stop wearing their masks once they put their masks on. In general, the proportions of all conditions and group transmissions were reduced with the introduction of social approval. And as the social penalty increase, the proportions of health conditions and group transmissions also received more decreases. Namely, when the social penalty became 30, the peak of the proportions of the exposed, infected, and dead agents dropped to 0.06, 0.06, and 0.014, which were nearly half of their original values in the basic model (Fig. 3(e)(f)). Similarly, the proportions of household transmissions and none-household transmissions both reduced by half. This could be contributed by the increase of social penalty, which led to more mask-wearings in the population. By wearing their masks, more agents were able to receive a 0.7 reduction (CDC, 2020) in their probabilities in being exposed with the virus when they interact with an infectious neighbor. And this could effectively reduce the proportions of conditions and transmissions from the groups.

3.3.2 Simulation 4: Social approvals and mask-wearings

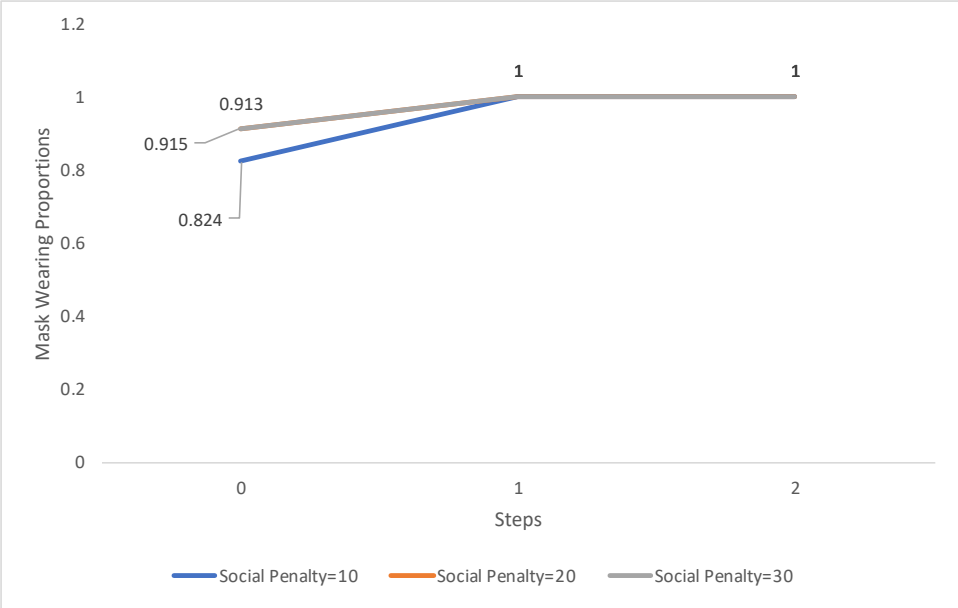
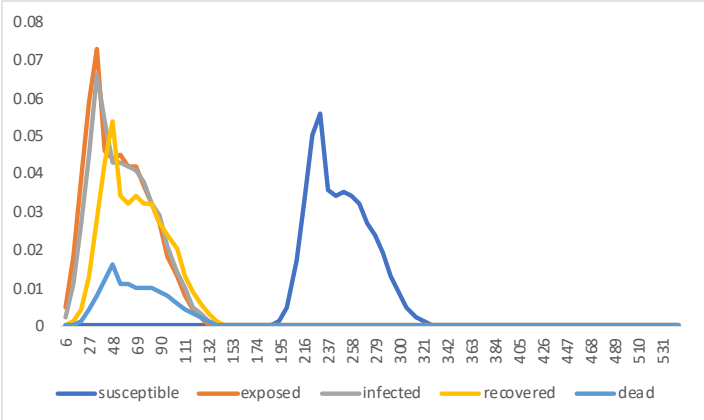


Fig. 4. This depicts the proportions of mask wearers in the population during a 3-steps interval.

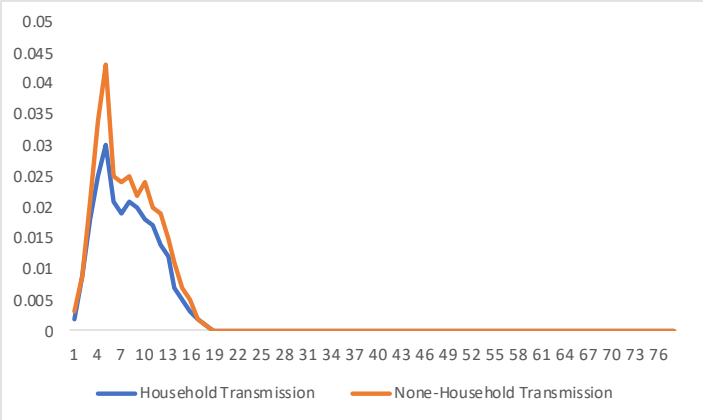
This simulation included the same settings as simulation 3, but it particularly studied the effects of social approval on the proportion of agents who wore masks after each step for a period of 3 steps. Without social approval, the agents maintained a 0.49 proportion of wearing masks during the period (Fig. 4), which was nearly consistent with the randomly assigned proportions at the beginning. However, when the social penalty was introduced, all cases had a substantial increase in mask-wearing just after the first step, and they all reached 1 after 1 step for their weak-wearing proportion. Namely, when the social penalty was set to 10, the agents showed a proportion of 0.824 of mask-wearing after the initial step. When the social penalty increased to 20, the proportion increased to 0.913. And when the social penalty became 30, the proportion further increased to 0.915. This suggests social approval does have effects in promoting public mask wearings. However, although the increased social penalties led to higher levels of mask wearing, the increase of the effects became much milder. For instance, after applying a social penalty of 10, there was a more than 68% increase in mask wearings after the first-step. But compares to the value of 10, when the social penalty was increased to 20, the increase in mask wearing only raised around 10.8%. In contrast to this 20, when the social penalty of 30 was applied, such increase became nearly 0.22%. This could be because an equilibrium or saturation point in public mask wearings was reached during the increase of the social penalties. As the result, a further increased social penalty beyond such value would only result in a smaller percentage increase of the mask wearings.

3.4 Simulation 5: Vaccinations

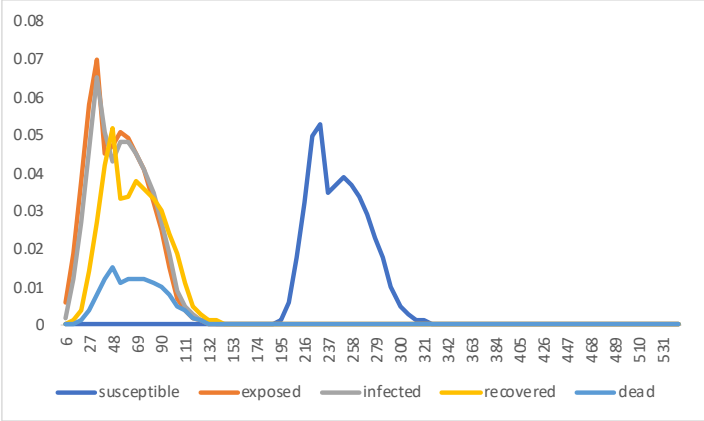
In simulation 5, it studied the effects of the different vaccine coverages and prioritizing the elder group population. In order to be “vaccinated," agents needed to complete two vaccine



(a) coverage=0.04, age restriction=false



(b) coverage=0.04, age restriction=false



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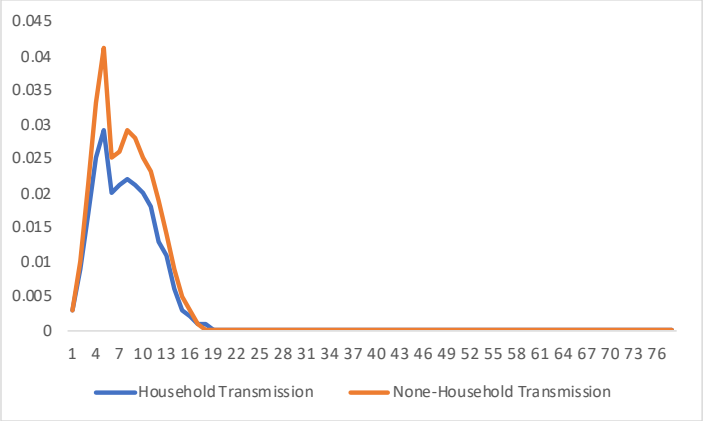
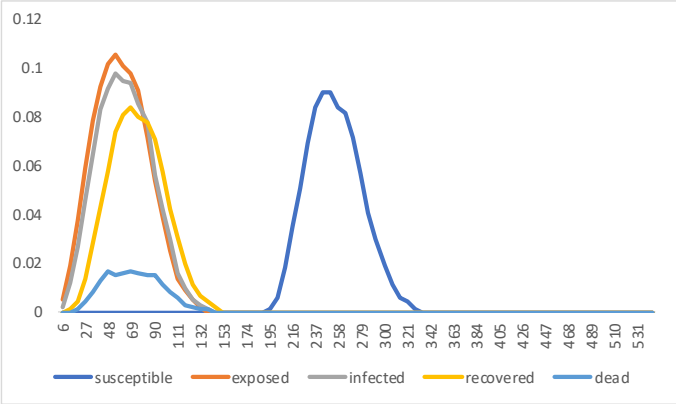


(d) coverage=0.1, age restriction=false

Fig. 5.1. The charts show the effects of the coverages of 0.04 and 0.1 on the health conditions and group transmissions under no age restriction.

doses, which is commonly used in the current vaccination practices (CDC, 2021). At each step, if the proportion of the vaccinated number against the population number was within the population proportion coverage, the agents who were not vaccinated could receive their dose. Make be more align to the real world situation, after an agent received its first dose, it needed to wait for an interval of 30 steps before it could receive its second dose (CDC, 2021). When the effects of "prioritizing the elder population" was studied, the agents' ages were also checked to determine if they were eligible for vaccinations. Once an agent became vaccinated, it needed to wait for an interval of 14 steps for the vaccine to take effects (CDC, 2021). When the countdown ended, it could no longer be infected. As the result, when under no age restriction, the vaccine coverage of 0.04 (CDC, 2021) led to a nearly 40% drop in proportions of all conditions (Fig. 5.1(a)). There was also a sharp decrease from 0.07 after around 40 steps. But when the coverage raised to 0.1, there was no substantial changes in the effects of reducing the transmission. Surprisingly, compares to the coverage of 0.04, the proportion of the exposed and infected conditions here received a small increase from 48th and 80th step. This could be due to the



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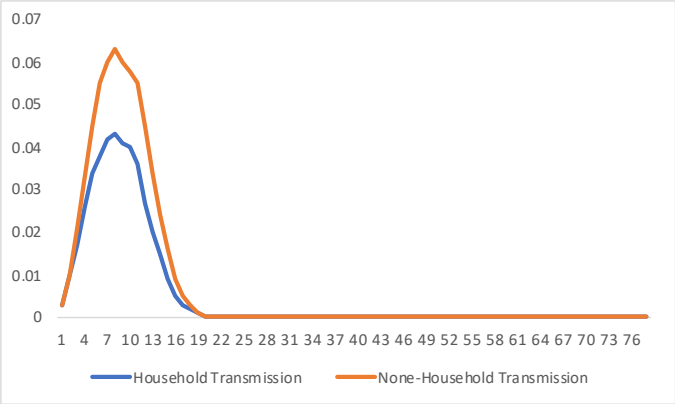
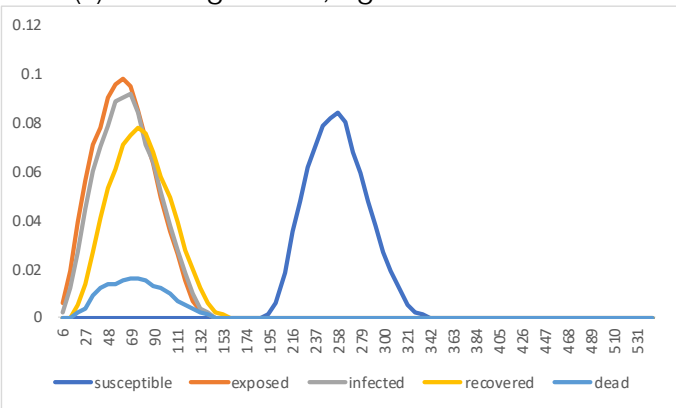
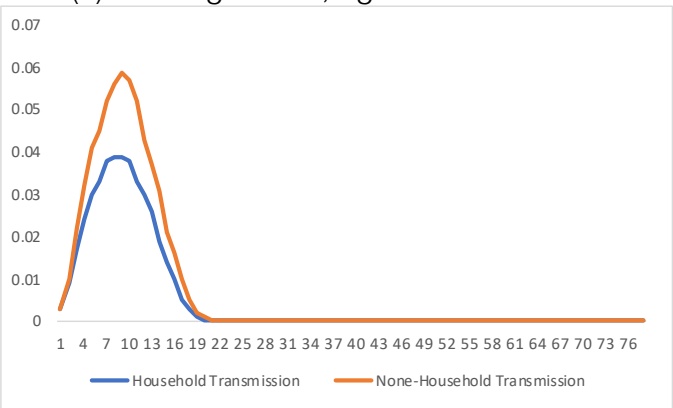


(b) coverage=0.04, age restriction=true



(c) coverage=0.1, age restriction=true



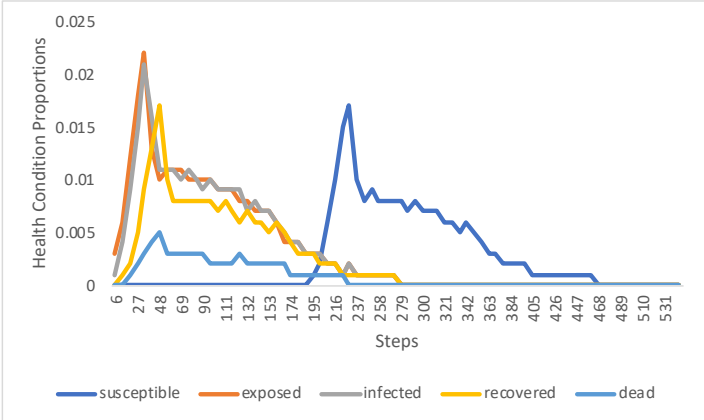
(d) coverage=0.1, age restriction=true

Fig. 5.2. The charts show the effects of the coverages of 0.04 and 0.1 on the health conditions and group transmissions under the age restriction.

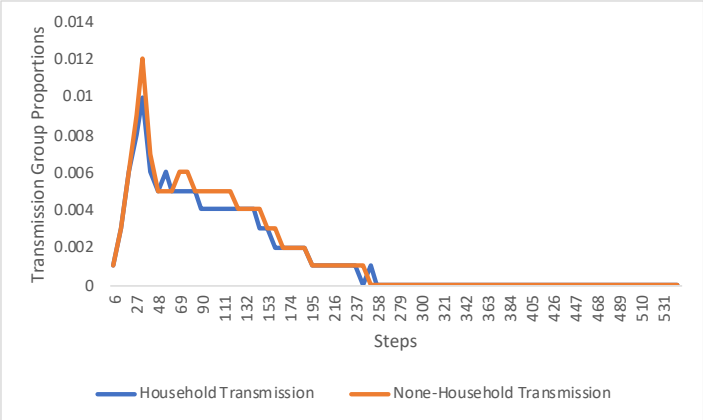
increased coverage led to a higher vaccinated number after 48 steps, which made the vaccinated proportion to quickly exceed the coverage. However, after applying the age restriction, although it still shows a decreased proportions for all conditions, its overall effects were smaller than the one without the restriction.

3.5 Simulation 6: Testing with more realistic settings

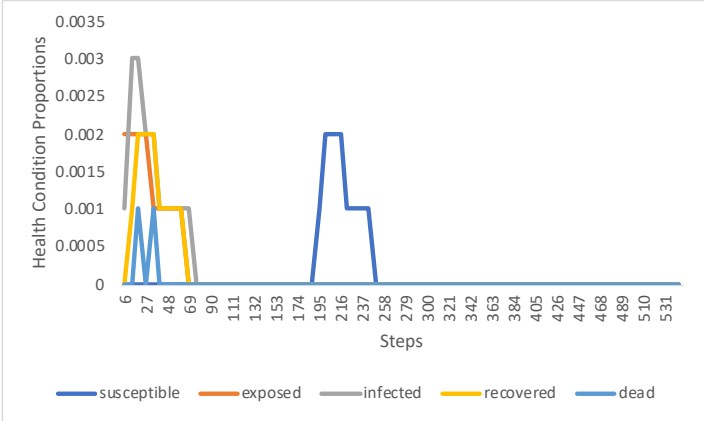
This simulation was identical to simulation 2, except that it incorporated different factors to better study the effects of testing under real-world situations. To make the setting more realistic, the previous studied factors of social approval, mutation, vaccination (with no age restriction) were included. Under the combined setting of these factors, the effects of testing was more throughly investigated. As the result, testing decreased all conditions by almost a 85% on average (Fig. 6). Furthermore, the proportions of household and none-household transmission both dropped to 0.001 from 0.012 at 27th step, and they all dropped to 0 after around 48 steps. Also, the proportions of all conditions and group transmissions reached 0 after 258 and 48 steps



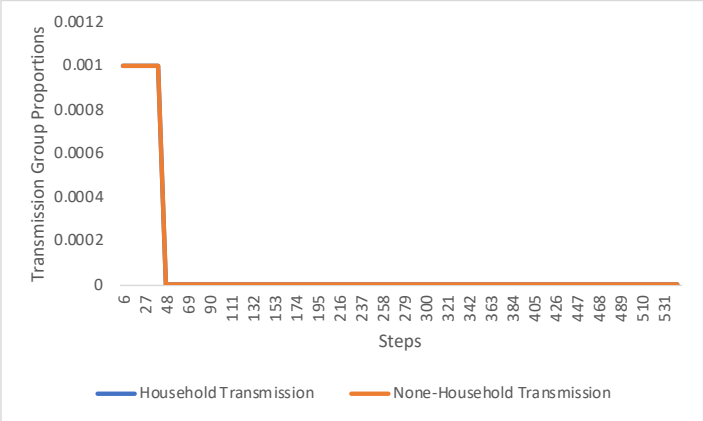
(a) testing=false



(b) testing=false



(c) testing=true



(d) testing=false

Fig. 6. The figures display the proportions of the health conditions and group transmissions when there was testing or no testing after incorporating the different factors, such as the social approvals, virus mutation, and vaccination.

respectively. This is much shorter than the ones of 468 and 258 of the trial without testing. These effects could be explained by that testing and quarantine allowed the population to quickly discover the infectious agents and isolate them, which led to more efficient control of the contamination spread.

4. Discussion

In this model, the effects of the different parameters, such as testing, social approvals, and vaccinations were investigated. By applying the space and mobility with the mobile agents, it allowed us to better study their effects in more realistic settings. In the model, it also showed that the transmission and the spread of the virus could be highly dependent on the different factors. By studying the combinations of these factors, it provided us with more insights into the

transmission of this virus. However, just like any model, there could be some limitations in the different aspects in this model. In order to better understand the spread of this pandemic under the different factors, more realistic assumptions and settings could be incorporate into the future research.

4.1 Testing frequencies and quarantine intervals

It was found in the simulation that testing and quarantine did have a substantial impact in controlling the contamination spread and improving the public health conditions. With the combination of the testing frequency of 7 steps and the quarantine length of 14 steps, it most effectively reduced the proportions of all health conditions and limited the spread. However, the trials with higher testing intervals were able to reduce the health condition proportions to 0 faster than the previous combination. Although this reduction in their time in restoring the population health conditions could be explained by that more infectious agents were able to be tested and start their quarantines at the same time, more combinations of testing intervals and quarantine lengths along with their transmission ending dates could be further studied in the future research. Because this model mainly looked at the increase of the testing frequencies and quarantine length with an increment of 7 steps, which corresponds to a week in the real-world length. But by studying the combinations of their smaller increments, it might allow us to more throughly explain the differences in the transmission ending dates for the different combinations. Furthermore, there could be limited testing access during the early stage of the pandemic (Manabe et al., 2020), so it'd also be interesting for the future models to incorporate this aspect into their research.

4.2 Transmission Groups

When the health conditions of the agents were checked at each step, their probabilities of being transmitted with the virus were highly dependent on the groups they belong to. Became if a agent and its infectious neighbor were both in the household group, it would correspond to a higher likelihood for a household transmission. Otherwise, it would receive a general level of the transmission risk. However, although the incorporation of household and none-household groups could create a simplicity and aligns with some of the current literatures (Fung et al., 2020), it might not best represent the real-world situations. Because in the real life, people have more than one group other than the household, such as school, religious, and work groups. And each of them have a different transmission probability. Therefore, it might be interesting for the future research to include more of these social network groups and the effects of their transmission probabilities under the different conditions.

4.3 Social approval baseline and social penalties

By including a baseline score of 70 and the social penalties of 10, 20, and 30, simulation 2 studied the effects of social approvals on the proportions of population health conditions and

mask wearings. It was found when the social penalty reaches its maximum of 30, there was a substantial decrease of 50% in all health conditions and group transmissions along with a 68% increase in mask wearings. This is also consistent with some of the current research findings (Eikenberry, 2020). But similar to that of testing interval and quarantine lengths, the increase of social penalty beyond the point of 10 led such effects to gradually become weaker (Fig. 3). However, the effects of the social approvals could be applied to explain why does the pandemic seemed harder to control in some countries than others. For instance, in some collectivist countries, the interests and needs of the group are more emphasized than that of the individuals (Sinha, D., & Tripathi, 1994). And maintaining the social harmony and fulfilling the social expectations could be one of the significant factors of gaining social acceptance. Under the collectivist culture, people might become more encouraged to wear masks to protect the interest of the group and avoid being shamed by the others for breaking the social norm (Madrim, 2020). And by applying the social approvals, these people in these countries were able to cooperate and respond to pandemic more efficiently (Jiang et al., 2021). However, the simulations here assumed that once the agents start to put on their masks, they would always have their mask on. But this isn't necessarily true in the real world, wherein some people would only wear masks in certain situations (such as being in a crowed area) and others would simply stop wearing masks after some time. Thus, the future research about the effects of mask wearing could take more of such factors into account. In addition, the effects of more varied social approval baselines and social penalties could also be investigated in the future research, which could allow us to further study the effects of social approvals in the control of the pandemic.

4.5 Vaccination eligibility and effectiveness length

Under no age restriction, simulation 5 found an around 40% drop in the proportions of all health conditions. However, after applied such restriction, although the transmission proportion was also reduced, the magnitude of such decrease was much weaker than the previous one. However, to make the simulation more realistic, future models could have allow the vaccinations to begin after a certain time. And when applying the age restriction, they could allow the other age groups to start to be vaccinated after a period of time when most of the elder agents have completed their vaccinations. Furthermore, this model assumes that the vaccines have the unlimited effectiveness length, which is not always true for many real-life vaccines. Thus, an effectiveness length of the vaccine could be incorporated into the future models. Additionally, the effects of the different vaccine coverages based on the population sizes could also be systematically studied in the future research.

4.6 Transmission factors in models of epidemiology

This model provided an analysis of the effects of the different factors in the Covid-19 transmission and population health conditions under the incorporations of space and mobility. The findings of the model could help to better understand and predict such effects under the different real-world situations. However, for more realistic models, more aspects and

combinations of the different factors could be considered. For the future directions, it would be interesting to investigate the additional questions of: How would the effectiveness of the different vaccines affect the transmission and population health conditions? How would the individual immunities interact with the transmission dynamics? What role does people's resources play during the pandemic? What other factors lead some people to be at higher transmission risks? By trying to answer such questions, we are able to revive more insights and understanding about the roles of the different factors in the transmission, which can also allows us to discover more about the underlying epidemiology mechanism of this pandemic and take better procedures in the future.

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